

OptiCrop : Smart Agricultural Production Optimization Engine

Abstract

OptiCrop – Machine Learning Based Agricultural Recommendation System is an intelligent web application designed to assist farmers and agricultural stakeholders in selecting the most suitable crop based on soil nutrients and environmental conditions. The system utilizes Machine Learning algorithms to analyze key agricultural parameters, including Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, Soil pH, and Rainfall. Multiple classification models were trained and evaluated, with the Random Forest classifier achieving the highest prediction accuracy of 99.32%. The trained model was integrated into a Flask-based web application that provides real-time crop recommendations along with prediction confidence, water requirements, suitable temperature range, and crop descriptions. The project demonstrates the practical application of Machine Learning in agriculture, enabling data-driven crop selection while promoting sustainable farming practices. Future enhancements include cloud deployment, weather API integration, fertilizer recommendations, crop disease detection, and mobile application support.

Project Description:

OptiCrop is a Machine Learning-powered web application that recommends the most suitable crop based on soil nutrients and environmental conditions. The system analyzes seven agricultural parameters—Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, Soil pH, and Rainfall—to predict the best crop for cultivation.

The application uses a supervised Machine Learning approach, where multiple classification algorithms including Logistic Regression, K-Nearest Neighbors, Decision Tree, and Random Forest were trained and evaluated. Based on comparative performance, the Random Forest classifier was selected as the final prediction model, achieving an accuracy of **99.32%**.

Developed using the Flask framework, the application provides a simple and interactive web interface where users can enter agricultural parameters and instantly receive crop recommendations along with the prediction confidence, water requirement, ideal temperature range, and crop description. The trained

Machine Learning model and feature scaler are serialized using Pickle and integrated with the Flask backend for real-time predictions.

OptiCrop aims to assist farmers, agricultural students, and researchers in making informed crop selection decisions, ultimately improving agricultural productivity and promoting sustainable farming practices.

Scenarios:

Scenario-1 Smart Crop Recommendation for Farmers

A farmer enters soil and environmental details such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH level, and rainfall. The system analyzes the data and recommends the most suitable crop for maximum yield and better farming efficiency.

Scenario 2: Crop Suitability and Environmental Assessment

A user wants to understand whether the current soil and climate conditions are suitable for a particular crop. The application evaluates the input parameters and provides insights about crop compatibility, environmental suitability, and productivity potential.

Scenario 3: Agricultural Research and Policy Planning

An agricultural researcher or policymaker uses the system to analyze crop-environment relationships and identify patterns in agricultural production. The platform helps in making data-driven decisions for sustainable farming strategies and resource optimization.

Technical Architecture

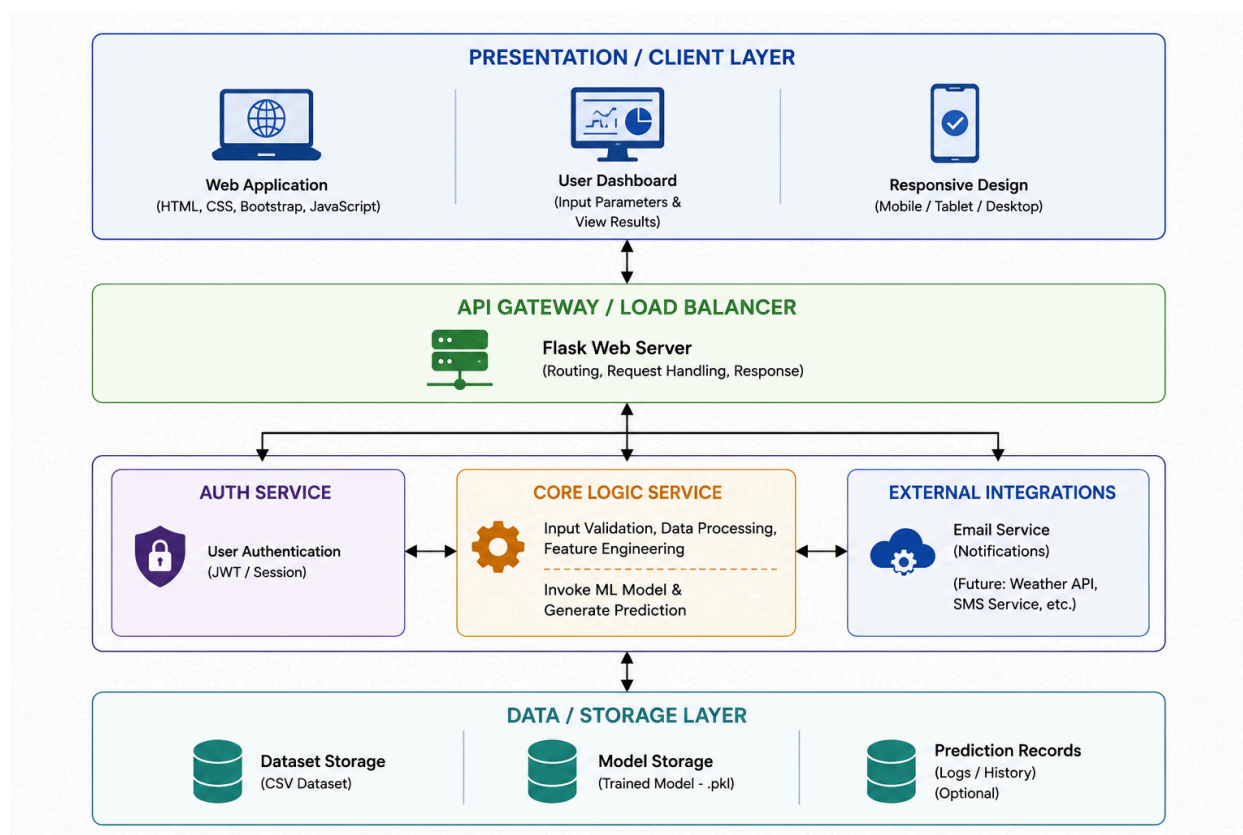


Figure 1: Technical Architecture of the OptiCrop System

Description

The OptiCrop system follows a three-layer architecture consisting of the Presentation Layer, Application Layer, and Machine Learning Layer. Each layer performs a specific function to provide accurate crop recommendations based on user inputs.

1. Presentation Layer (Frontend)

The Presentation Layer is responsible for user interaction. It is developed using HTML, CSS, Bootstrap, and JavaScript, providing a responsive and user-friendly interface. Users enter agricultural parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, Soil pH, and Rainfall through the web application.

2. Application Layer (Flask Backend)

The Application Layer is implemented using the Flask framework. It receives input data from the frontend, performs server-side validation, preprocesses the data using the saved feature scaler, and communicates with the trained Machine Learning model.

The backend is also responsible for:

- Processing prediction requests
- Loading the trained Pickle model
- Returning prediction results in JSON format
- Handling errors and invalid inputs

3. Machine Learning Layer

The Machine Learning Layer contains the trained Random Forest Classification Model and the StandardScaler used during preprocessing.

The workflow of this layer includes:

- Receiving scaled feature values
- Predicting the most suitable crop
- Calculating prediction confidence
- Returning the predicted crop label to the Flask application

The trained model achieved an overall prediction accuracy of 99.32%, making it highly reliable for crop recommendation.

4. Crop Information Module

After obtaining the prediction, the system retrieves additional crop details from the crop information module.

This module provides:

- Crop name
- Water requirement

- Recommended temperature range
- Brief crop description
- Emoji/icon representation

The information is displayed on the user interface along with the prediction result to help users better understand the recommendation.

5. User Output

Finally, the application displays:

- Recommended Crop
- Prediction Confidence
- Water Requirement
- Suitable Temperature Range
- Crop Description

This provides users with a complete recommendation instead of only displaying the predicted crop name.

Project Workflow

The OptiCrop system follows a structured workflow that transforms user-provided agricultural parameters into accurate crop recommendations using Machine Learning. The workflow consists of data collection, preprocessing, model prediction, and result visualization.

The overall workflow of the system is illustrated below.

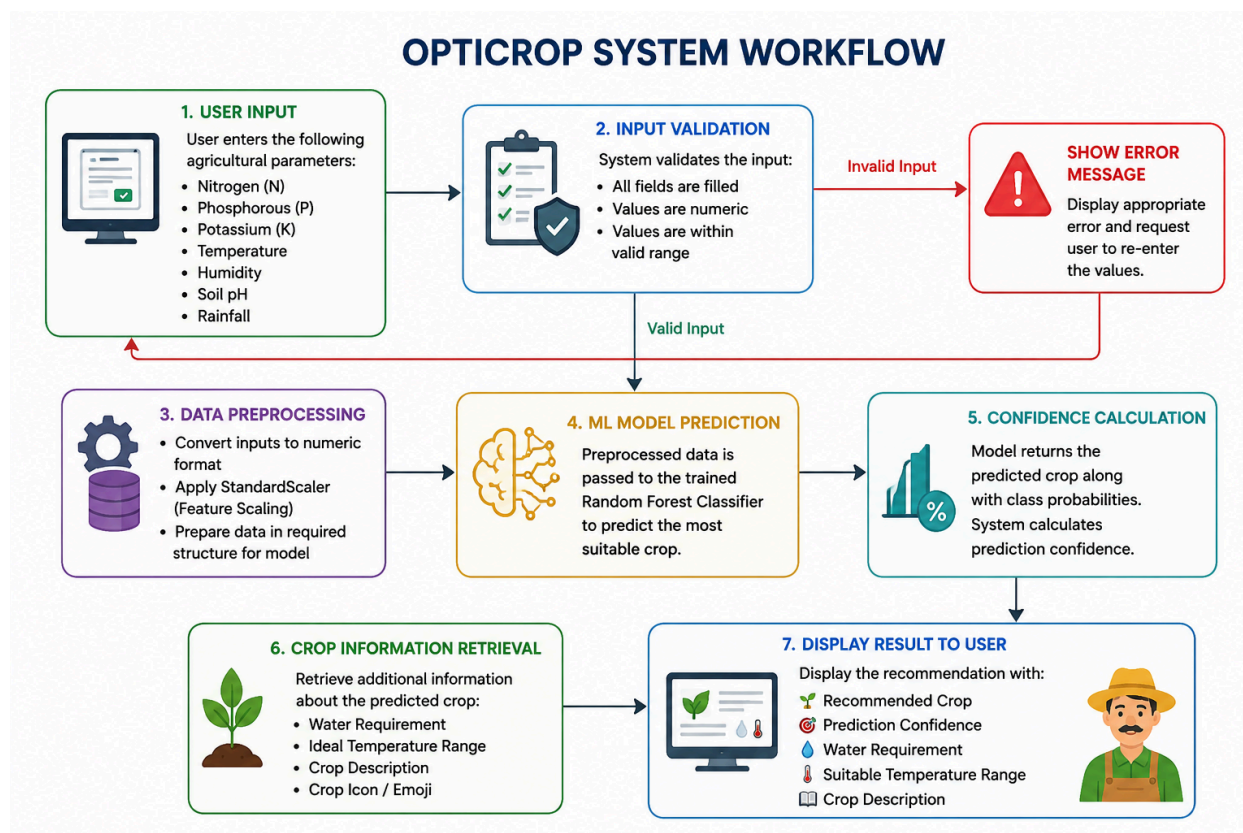


Figure 2: Workflow of the OptiCrop System

Step 1: User Input

The workflow begins when the user accesses the OptiCrop web application. The user enters the required agricultural parameters through the prediction form.

The input parameters include:

- Nitrogen (N)
- Phosphorous (P)
- Potassium (K)
- Temperature
- Humidity
- Soil pH
- Rainfall

These parameters represent the soil nutrient levels and environmental conditions required for crop recommendation.

Step 2: Input Validation

Once the user submits the form, the application performs validation on both the client side and the server side.

The validation process ensures that:

- All required fields are filled.
- Numerical values are entered.
- Inputs fall within valid agricultural ranges.
- Invalid or incomplete data is rejected with appropriate error messages.

This improves system reliability and prevents incorrect predictions.

Step 3: Data Preprocessing

After successful validation, the backend preprocesses the input data.

The preprocessing stage includes:

- Conversion of user inputs into numerical format.
- Feature scaling using the previously trained **StandardScaler**.
- Formatting the data into the structure expected by the Machine Learning model.

Using the same scaler that was applied during model training ensures prediction consistency.

Step 4: Machine Learning Prediction

The preprocessed data is passed to the trained **Random Forest Classifier**.

The model analyzes all seven input parameters and predicts the most suitable crop from the available **22 crop categories**.

Additionally, the model calculates the prediction confidence using class probabilities.

Step 5: Crop Information Retrieval

Once the crop is predicted, the application retrieves additional information from the crop information module.

The retrieved details include:

- Crop Name
- Confidence Score
- Water Requirement
- Ideal Temperature Range
- Brief Crop Description

This provides users with meaningful information instead of only displaying the crop name.

Step 6: Display Results

Finally, the Flask backend sends the prediction results to the frontend.

The application displays:

-  Recommended Crop
-  Prediction Confidence
-  Water Requirement
-  Suitable Temperature Range
-  Crop Description

The user can reset the form and perform multiple predictions without restarting the application.

Milestone 1: Project Setup and Dataset Preparation

Objective

The primary objective of this milestone was to establish the development environment, collect the required dataset, and prepare the data for Machine Learning model development. This phase forms the foundation of the OptiCrop system by ensuring that the dataset is clean, structured, and ready for training and evaluation.

Activity 1.1: Development Environment Setup

Objective

To configure the software environment required for developing and testing the OptiCrop application.

Tools Used

Tool	Purpose
Python 3.11	Programming Language
Visual Studio Code	IDE
Flask	Backend Framework
Git	Version Control
GitHub	Source Code Management

Steps Performed

- Installed Python and configured the virtual environment.
- Installed required Python libraries using `requirements.txt`.
- Configured Flask project structure.
- Created folders for data, models, templates, and static files.
- Initialized Git repository and connected it to GitHub.
- Verified successful execution of the Flask application.

Activity 1.2: Dataset Collection

Objective

To obtain a reliable dataset for training the crop recommendation model.

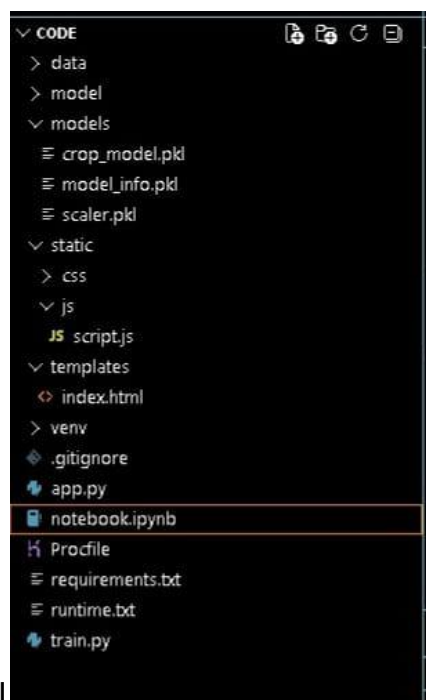
Dataset Information

Property	Value
Dataset Name	Crop Recommendation Dataset
Source	Kaggle
Records	2200
Features	7
Target Classes	22 Crops

Input Features

- Nitrogen (N)
- Phosphorous (P)
- Potassium (K)
- Temperature
- Humidity
- Soil pH
- Rainfall

Output



- Crop Label

Activity 1.3: Data Exploration and Analysis

Objective

To understand the characteristics of the dataset before preprocessing and model training.

Activities Performed

- Loaded the dataset using Pandas.
- Verified data types.
- Checked dataset dimensions.
- Confirmed absence of missing values.
- Examined feature distributions.
- Identified target crop classes.

Dataset Summary

Parameter	Value
Total Samples	2200

Total Features	7
Crop Categories	22
Missing Values	0
Duplicate Records	0

```
print(df.shape)
```

Python

... (2200, 8)

```
df.head()
```

Python

	N	P	K	temperature	humidity	ph	rainfall	label
0	90	42	43	20.879744	82.002744	6.502985	202.935536	rice
1	85	58	41	21.770462	80.319644	7.038096	226.655537	rice
2	60	55	44	23.004459	82.320763	7.840207	263.964248	rice
3	74	35	40	26.491096	80.158363	6.980401	242.864034	rice
4	78	42	42	20.130175	81.604873	7.628473	262.717340	rice

Milestone 2: Data Preprocessing and Machine Learning Model Development

Objective

The objective of this milestone was to preprocess the collected agricultural dataset, prepare it for machine learning, train multiple classification algorithms, evaluate their performance, and select the most accurate model for crop recommendation.

Activity 2.1: Data Preprocessing

Objective

To prepare the dataset for training by handling missing values, selecting relevant features, and standardizing the input data.

Steps Performed

- Loaded the dataset using the Pandas library.
- Checked for missing values in all columns.
- Replaced missing values (if any) using the median of the respective feature.
- Selected seven agricultural parameters as input features.
- Selected the crop label as the target variable.
- Applied feature scaling using **StandardScaler**.
- Divided the dataset into training and testing sets using an **80:20 split**.

Input Features

Feature	Description
Nitrogen (N)	Soil nitrogen content
Phosphorous (P)	Soil phosphorous content
Potassium (K)	Soil potassium content
Temperature	Ambient temperature (°C)
Humidity	Relative humidity (%)
Soil pH	Soil acidity/alkalinity
Rainfall	Rainfall (mm)

Target Variable

- Crop Label (22 Crop Categories)

Activity 2.2: Feature Scaling

Objective

To normalize the numerical feature values before training the machine learning models.

Method Used

- StandardScaler

Purpose

Feature scaling ensures that all numerical features contribute equally during model training and improves the convergence of algorithms such as Logistic Regression and K-Nearest Neighbors.

Activity 2.3: Machine Learning Model Training

Objective

To train multiple supervised machine learning algorithms and compare their performance.

Algorithms Evaluated

Algorithm	Purpose
Logistic Regression	Baseline linear classifier
K-Nearest Neighbors (KNN)	Distance-based classification
Decision Tree	Rule-based classification
Random Forest	Ensemble learning classifier

Each model was trained using the standardized training dataset and evaluated on the testing dataset.

Activity 2.4: Model Performance Comparison

The performance of all trained models was compared using prediction accuracy.

Machine Learning Model	Test Accuracy
Logistic Regression	96.36%
K-Nearest Neighbors	95.68%
Decision Tree	98.64%
Random Forest	99.32%

```
===== MODEL COMPARISON =====
```

```
Logistic Regression: 96.36%  
K-Nearest Neighbors: 95.68%  
Decision Tree: 98.64%  
Random Forest: 99.32%
```

```
Selected Model : Random Forest  
Accuracy : 99.32%
```

The Random Forest Classifier achieved the highest prediction accuracy and was selected as the final model for deployment in the OptiCrop application.

Activity 2.5: Model Evaluation

The selected Random Forest model was evaluated using a classification report consisting of:

- Precision
- Recall
- F1-Score
- Accuracy

Classification Report				
	precision	recall	f1-score	support
apple	1.00	1.00	1.00	23
banana	1.00	1.00	1.00	21
blackgram	1.00	1.00	1.00	20
chickpea	1.00	1.00	1.00	26
coconut	1.00	1.00	1.00	27
coffee	1.00	1.00	1.00	17
cotton	1.00	1.00	1.00	17
grapes	1.00	1.00	1.00	14
jute	0.92	1.00	0.96	23
kidneybeans	1.00	1.00	1.00	20
lentil	0.92	1.00	0.96	11
maize	1.00	1.00	1.00	21
mango	1.00	1.00	1.00	19
mothbeans	1.00	0.96	0.98	24
mungbean	1.00	1.00	1.00	19
muskmelon	1.00	1.00	1.00	17
orange	1.00	1.00	1.00	14
papaya	1.00	1.00	1.00	23
pigeonpeas	1.00	1.00	1.00	23
pomegranate	1.00	1.00	1.00	23
rice	1.00	0.89	0.94	19
watermelon	1.00	1.00	1.00	19
accuracy			0.99	440
macro avg	0.99	0.99	0.99	440
weighted avg	0.99	0.99	0.99	440

Evaluation Result

Metric	Result
Overall Accuracy	99.32%
Precision	High across all crop classes
Recall	High across all crop classes
F1-Score	Approximately 0.99

The evaluation demonstrated that the selected model provides highly accurate crop recommendations across all 22 crop categories.

Activity 2.6: Model Serialization

Objective

To save the trained model and preprocessing objects for future predictions.

Files Generated

File	Purpose
<code>crop_model.pkl</code>	Stores the trained Random Forest model
<code>scaler.pkl</code>	Stores the trained StandardScaler object

The **Pickle** library was used to serialize both files. During application execution, these files are loaded into memory, eliminating the need to retrain the model each time the application starts.

Activity 2.7: Unsupervised Learning Analysis

Although the primary objective of the project is supervised crop classification, an additional **K-Means Clustering** experiment was performed to explore the distribution of agricultural data.

The clustering process helped analyze similarities among input samples but was **not used** for the final prediction system.

Milestone 3: Web Application Development and System Integration

Objective

The objective of this milestone was to develop a responsive web application that enables users to interact with the trained Machine Learning model. The Flask framework was used to integrate the backend with the frontend, allowing real-time crop recommendations based on user-provided agricultural parameters.

Activity 3.1: Flask Application Development

Objective

To develop the backend server responsible for processing user requests and communicating with the trained Machine Learning model.

Technologies Used

Technology	Purpose
Flask	Backend Web Framework
Python	Backend Programming Language
Pickle	Loading trained model and scaler

Activities Performed

- Created the Flask application (`app.py`).
- Configured routing for the home page and prediction endpoint.
- Loaded the trained Random Forest model using Pickle.
- Loaded the StandardScaler object for preprocessing.
- Configured JSON request and response handling.
- Implemented exception handling for prediction errors..



```

(venv) PS C:\Users\Dharshan\Documents\OptiCrop\5.Project Development Phase\code> python app.py
(venv) PS C:\Users\Dharshan\Documents\OptiCrop\5.Project Development Phase\code> python app.py
Model Loaded Successfully
{'model_name': 'Random Forest', 'requires_scaling': False}
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.40.185.248:5000
Press CTRL+C to quit
  
```

Activity 3.2: Frontend Development

Objective

To design a simple, responsive, and user-friendly interface for entering agricultural parameters.

Technologies Used

Technology	Purpose
HTML5	Page Structure
CSS3	User Interface Styling
Bootstrap 5	Responsive Design
JavaScript	Client-side Validation and API Requests

Features Implemented

- Responsive input form
- Seven agricultural input fields
- Predict button
- Reset button
- Dynamic result display
- Error message display

Activity 3.3: Machine Learning Model Integration

Objective

To integrate the trained Random Forest model into the Flask application.

Integration Process

The following steps were performed:

1. Loaded the serialized Random Forest model (`crop_model.pkl`).
2. Loaded the trained StandardScaler (`scaler.pkl`).
3. Received user input from the frontend.
4. Applied feature scaling.
5. Generated crop prediction.
6. Calculated prediction confidence.
7. Returned prediction results to the frontend.

Activity 3.4: Input Validation

Objective

To ensure that only valid agricultural parameters are processed by the system.

Validation Techniques

Client-Side Validation

Implemented using JavaScript.

Validation checks include:

- Mandatory input fields
- Numeric values only
- Acceptable agricultural ranges
- User-friendly error messages

Server-Side Validation

Implemented using Flask.

Validation checks include:

- Data type verification
- Range validation
- Invalid request handling
- Exception management

Activity 3.5: Prediction Result Generation

Objective

To present meaningful prediction results to the user.

Prediction Output

After processing the input values, the application displays:

- Recommended Crop

- Prediction Confidence (%)
- Water Requirement
- Ideal Temperature Range
- Crop Description

The prediction confidence is calculated using the probability scores generated by the Random Forest classifier.

Activity 3.6: Crop Information Module

Objective

To provide additional information about the predicted crop.

Information Displayed

Information	Description
Crop Name	Predicted crop
Water Requirement	Low / Medium / High
Temperature Range	Suitable growing temperature
Crop Description	Brief overview of the crop
Crop Emoji	Visual representation of the crop

The crop information is retrieved from a predefined Python dictionary (`crop_info.py`) based on the predicted crop.

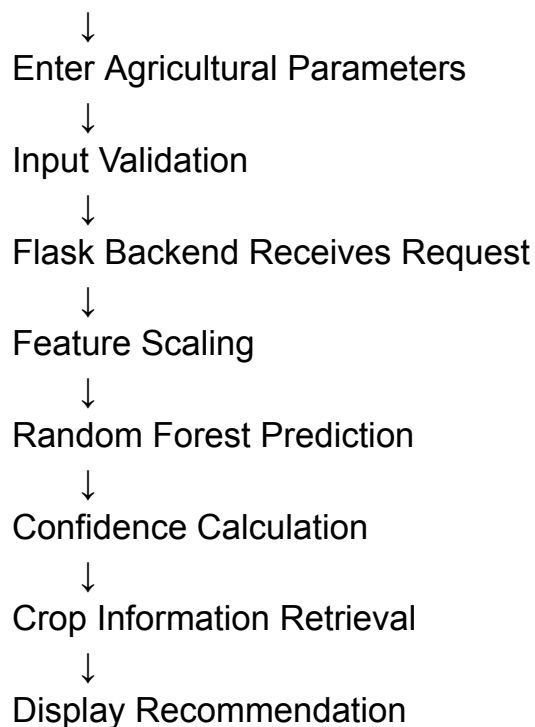
Activity 3.7: System Integration

Objective

To integrate the frontend, backend, and Machine Learning model into a complete working application.

System Workflow

User Opens Application



Milestone 4: Testing and Performance Evaluation

Objective

The objective of this milestone was to evaluate the functionality, reliability, accuracy, and performance of the OptiCrop application. Various testing techniques were performed to verify that each module of the system operated correctly and that the Machine Learning model produced accurate crop recommendations.

Activity 4.1: Functional Testing

Objective

To verify that all major functionalities of the application operate as expected.

Functional Test Cases

Test Case	Description	Expected Result	Status
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Home Page Loading	Verify application homepage loads successfully	Home page displayed correctly	Passed
User Input	Verify users can enter agricultural parameters	Input accepted successfully	Passed
Prediction Request	Verify prediction request reaches backend	Backend processes request successfully	Passed
Crop Recommendation	Verify recommended crop is displayed	Correct crop recommendation shown	Passed
Reset Function	Verify reset button clears all inputs	Form resets successfully	Passed

Activity 4.2: Input Validation Testing

Objective

To ensure that invalid or incomplete user inputs are properly handled.

Validation Scenarios

Validation	Expected Behaviour	Status
Empty Fields	Display validation message	Passed
Negative Values	Reject invalid input	Passed
Out-of-Range Values	Display error message	Passed
Non-Numeric Input	Prevent submission	Passed
Missing Parameters	Server rejects invalid request	Passed

Activity 4.3: Machine Learning Model Evaluation

Objective

To evaluate the prediction performance of the trained Machine Learning model.

Models Compared

Machine Learning Model	Accuracy
Logistic Regression	96.36%
K-Nearest Neighbors	95.68%
Decision Tree	98.64%
Random Forest	99.32%

The Random Forest classifier produced the highest prediction accuracy and was selected for deployment.

Activity 4.4: Performance Testing

Objective

To measure the application's response time and stability.

Performance Results

Parameter	Result
Average Prediction Time	Less than 1 second
Model Loading	Successful
Response Generation	Immediate
Multiple Consecutive Predictions	Successfully Executed
Application Stability	Stable



Activity 4.5: User Acceptance Testing (UAT)

Objective

To verify that the developed application satisfies the intended user requirements.

UAT Results

Requirement	Result
User-Friendly Interface	Passed
Accurate Crop Recommendation	Passed
Input Validation	Passed
Prediction Confidence Display	Passed
Crop Information Display	Passed

Activity 4.6: Test Result Summary

Overall Testing Summary

Testing Type	Result
Functional Testing	Passed
Input Validation Testing	Passed
Machine Learning Evaluation	Passed
Performance Testing	Passed
User Acceptance Testing	Passed

Overall Accuracy

Random Forest Classification Accuracy: 99.32%

Activity 4.7: Issues Identified and Resolved

During testing, a few implementation issues were identified and resolved.

Issue	Resolution
Model integration with Flask	Successfully integrated using Pickle serialization.
Invalid user input handling	Added client-side and server-side validation.
Prediction response formatting	Improved result display with confidence score and crop details.
Model loading efficiency	Loaded the model once during application startup for improved performance.

Milestone 5: Project Demonstration and Conclusion

This milestone focuses on demonstrating the completed OptiCrop application, verifying that all proposed functionalities work as expected, and presenting the final outcomes of

the project. The application was tested through multiple prediction scenarios to validate the performance of the Machine Learning model and the overall system workflow.

Activity 5.1: Application Deployment and Local Execution

Objective

To execute the OptiCrop application in the local development environment and verify that all components function correctly.

Steps Performed

- Activated the Python virtual environment.
- Installed all required dependencies using `requirements.txt`.
- Started the Flask development server.
- Opened the application using the local host URL.
- Verified successful loading of the home page.
- Confirmed successful loading of the trained Machine Learning model and StandardScaler.

Local Execution Command **`python app.py`**

Local URL **`http://127.0.0.1:5000`**

Activity 5.2: Demonstration of the Web Application

The OptiCrop application was demonstrated by entering agricultural parameters through the web interface. The application processed the user inputs, applied feature scaling, performed Machine Learning prediction, and displayed the recommended crop along with additional crop information.

Demonstrated Features

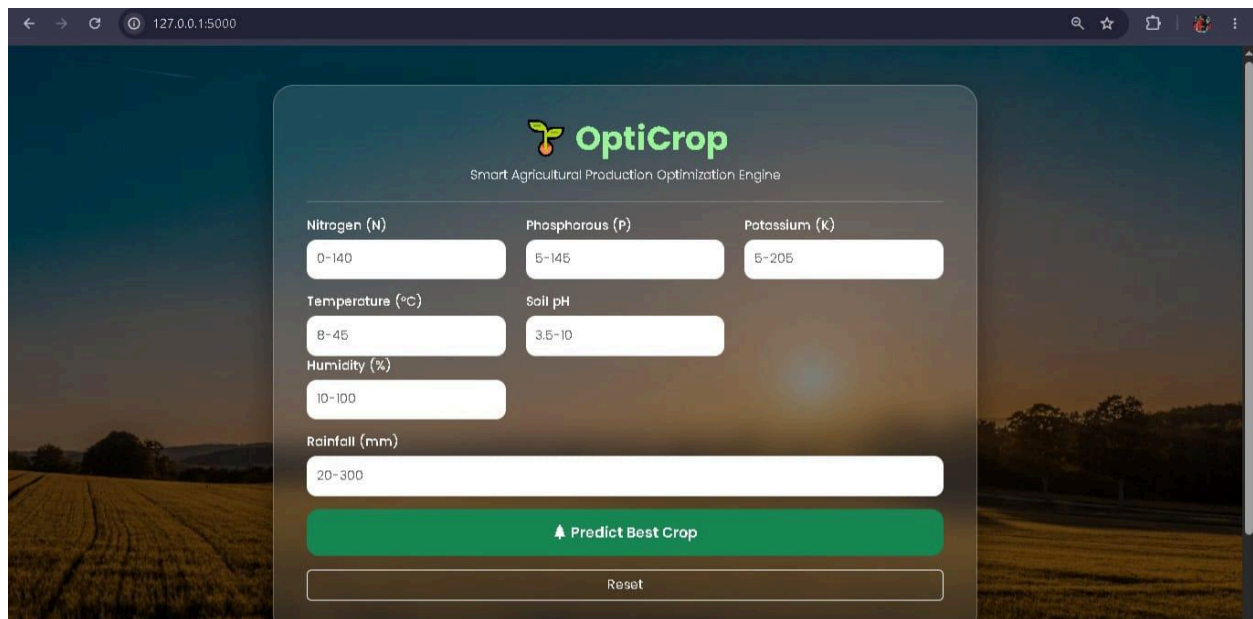
- Crop Recommendation Prediction
- Real-Time Machine Learning Prediction
- Prediction Confidence Display
- Water Requirement Information
- Suitable Temperature Range
- Crop Description
- Client-Side Validation

- Server-Side Validation
- Reset Functionality

Activity 5.3: Exploring the Website

Home Page

The Home Page serves as the primary interface of the OptiCrop application. It contains an intuitive prediction form where users can enter soil nutrient values and environmental parameters required for crop recommendation. The interface is designed using HTML, CSS, Bootstrap, and JavaScript to provide a clean, responsive, and user-friendly experience.

The screenshot shows a web browser window displaying the OptiCrop application. The browser's address bar shows the URL 127.0.0.1:5000. The page features a dark background with a blurred image of a field. In the center, there is a white card with the OptiCrop logo and the tagline "Smart Agricultural Production Optimization Engine". Below the logo, there are input fields for various agricultural parameters: Nitrogen (N) with a range of 0-140, Phosphorous (P) with a range of 5-145, Potassium (K) with a range of 5-205, Temperature (°C) with a range of 8-45, Soil pH with a range of 3.5-10, Humidity (%) with a range of 10-100, and Rainfall (mm) with a range of 20-300. A green button labeled "Predict Best Crop" is positioned below the input fields, and a "Reset" button is located at the bottom of the card.

Prediction Form

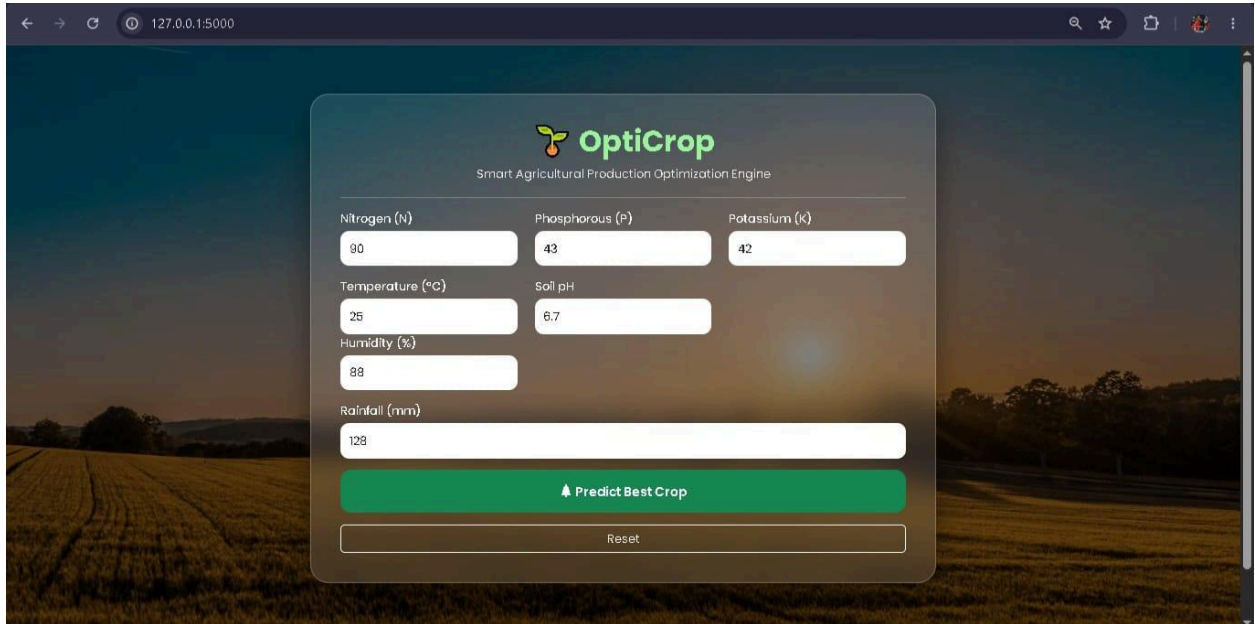
Description

The prediction form accepts the following agricultural parameters:

- Nitrogen (N)
- Phosphorous (P)
- Potassium (K)
- Temperature
- Humidity

- Soil pH
- Rainfall

After entering valid values, users can click the Predict Crop button to generate recommendations.



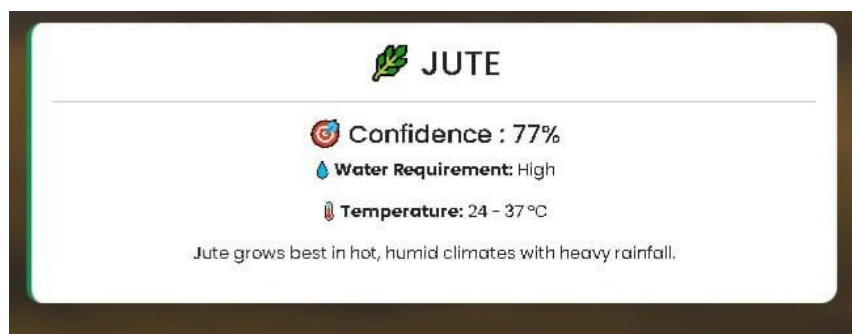
Prediction Result Page

Description

Once the prediction is completed, the application displays:

- Recommended Crop
- Prediction Confidence
- Water Requirement
- Suitable Temperature Range
- Crop Description

These results are generated dynamically using the trained Random Forest model integrated with the Flask backend.



Activity 5.4: Project Achievements

The OptiCrop project successfully achieved the objectives defined during the planning phase.

Key Achievements

- Developed a complete Machine Learning-based crop recommendation system.
- Trained and evaluated multiple classification algorithms.
- Achieved 99.32% prediction accuracy using the Random Forest classifier.
- Developed a responsive Flask web application.
- Implemented both client-side and server-side validation.
- Integrated Machine Learning predictions into a real-time web interface.
- Successfully demonstrated all major functionalities.

Activity 5.5: Advantages of the System

The developed application provides several advantages.

- High prediction accuracy using Machine Learning.
- Simple and user-friendly web interface.
- Fast prediction response time.
- Real-time crop recommendation.
- Helps farmers make informed crop selection decisions.
- Supports sustainable agricultural practices.
- Easy to extend with additional agricultural features.

Activity 5.6: Limitations

Although the application performs effectively, a few limitations remain.

- Uses a static dataset for training.
- No real-time weather data integration.
- No fertilizer recommendation module.
- No user authentication or login system.
- Currently runs as a local Flask application.
- Crop images are not integrated.

Activity 5.7: Future Enhancements

Several enhancements can further improve the OptiCrop system.

- Cloud deployment for public access.
- Integration of real-time weather APIs.
- Fertilizer recommendation system.
- Soil health analysis.
- Crop disease detection using Deep Learning.
- Android and iOS mobile applications.
- Multilingual support for farmers.
- IoT sensor integration for automatic data collection.

Conclusion

OptiCrop is a Machine Learning-based agricultural recommendation system developed to assist farmers and agricultural stakeholders in selecting the most suitable crop based on soil nutrients and environmental conditions. The application integrates a Random Forest Classification model with a Flask-based web interface to deliver accurate, real-time crop recommendations.

Multiple Machine Learning algorithms were trained and evaluated, with the Random Forest classifier achieving an overall prediction accuracy of 99.32%. The application successfully integrates data preprocessing, feature scaling, model prediction, and result visualization into a single user-friendly platform. In addition to predicting the most suitable crop, the system provides confidence scores, water requirements, suitable temperature ranges, and descriptive crop information to support informed agricultural decision-making.

The successful completion of this project demonstrates the practical application of Machine Learning in agriculture and highlights how intelligent decision-support systems can contribute to improved productivity and sustainable farming practices. With future enhancements such as cloud deployment, weather API integration, fertilizer

recommendations, and mobile accessibility, OptiCrop has the potential to evolve into a comprehensive smart farming solution.

Appendix

Source Code

Include the complete source code of the project.

OptiCrop/

├── app.py

├── train.py

├── requirements.txt

├── data/

├── models/

├── static/

├── templates/

└── notebook.ipynb

Dataset

Dataset Name: Crop Recommendation Dataset

Source: Kaggle

References

1. Crop Recommendation Dataset, Kaggle.
2. Flask Documentation – <https://flask.palletsprojects.com/>
3. Scikit-learn Documentation – <https://scikit-learn.org/>
4. Pandas Documentation – <https://pandas.pydata.org/>
5. NumPy Documentation – <https://numpy.org/>
6. Bootstrap Documentation – <https://getbootstrap.com/>
7. Python Documentation – <https://docs.python.org/>